Introduction (5%): Please provide a high-level introduction to your report. You can mention the most important findings and the overall organization of this report.

這次主要是在做 cartpole 和 pong 兩個遊戲的 DQN，其中也練習到了 PER, n-step, DDQN 等等的技巧，發現收斂快了很多。另外，我也嘗試了 Dueling DQN。

- Your implementation (20%): Please briefly explain both the concepts and your implementation details for Tasks 1–3. For each item below, you should first describe how the method works (e.g., how to compute Bellman error) and then explain how you implemented it in your code. Specifically, please describe:

  - How do you obtain the Bellman error for DQN?

計算當前神經網路 q 值預測值和目標值的差異，並套用公式

$$L_{\mathrm{DQN}}(\theta) := \frac{1}{2} \sum_{(s,a,r,s') \in D} \left(r + \gamma \max_{a' \in \mathcal{A}} Q(s', a'; \bar{\theta}) - Q(s, a; \theta) \right)^2,$$

```
with torch.no_grad():
    next_q_values = self.target_net(next_states).max(1)[0]
    target_q_values = rewards + self.gamma * next_q_values * (1 - dones)
loss = nn.MSELoss()(target_q_values, q_values)
```

  - How do you modify DQN to Double DQN?

使用 main 網路選擇最佳的動作，再使用 target 網路評估該動作的 q 值。把上一個截圖改成下圖：

```
with torch.no_grad():
    best_actions = self.q_net(next_states).argmax(1, keepdim=True)
    next_q_values = self.target_net(next_states).gather(1, best_actions).squeeze(1)
    target_q_values = rewards + (self.gamma ** self.n_step_n) * next_q_values * (1 - dones)
```

  - How do you implement the memory buffer for PER?

先實作出 sumtree 的架構提升訓練速度

```python
class SumTree:
    def __init__(self, capacity):
        self.capacity = capacity
        self.tree = np.zeros(capacity * 2 - 1)
        self.data = np.zeros(capacity, dtype=object)
        self.write_idx = 0

    def _propagate(self, idx, change):
        parent_idx = (idx - 1) // 2
        self.tree[parent_idx] += change
        if parent_idx != 0:
            self._propagate(parent_idx, change)

    def update(self, idx, p):
        change = p - self.tree[idx]
        self.tree[idx] = p
        self._propagate(idx, change)

    def add(self, p, data):
        self.data[self.write_idx] = data
        tree_idx = self.write_idx + (self.capacity - 1)
        self.update(tree_idx, p)
        self.write_idx = (self.write_idx + 1) % self.capacity
```

再完成 PrioritizedReplayBuffer 的實作，根據樣本的重要性，調整被抽樣到的機率。並增加權重調整偏差。

```python
def add(self, transition, error):
    ########## YOUR CODE HERE (for Task 3) ##########
    self.max_priority = max(self.max_priority, error)
    self.tree.add(error, transition)
    self.count = min(self.capacity, self.count + 1)
    ########## END OF YOUR CODE (for Task 3) ##########
    return

def sample(self, batch_size):
    ########## YOUR CODE HERE (for Task 3) ##########
    data_batch, leaf_idxs, priorities = [], [], []
    segment = self.tree.total_priority / batch_size
    self.beta = np.min([1., self.beta + self.beta_increment_per_sampling])

    for i in range(batch_size):
        v = random.uniform(i * segment, (i + 1) * segment)
        leaf_idx, p, data = self.tree.get_leaf(v)
        data_batch.append(data)
        leaf_idxs.append(leaf_idx)
        priorities.append(p)

    probability = np.array(priorities) / (self.tree.total_priority + 1e-10)
    weights = (1 / (self.tree.capacity * probability)) ** self.beta
    weights /= (weights.max() + 1e-10)
    ########## END OF YOUR CODE (for Task 3) ##########
    return list(zip(*data_batch)), leaf_idxs, weights
```

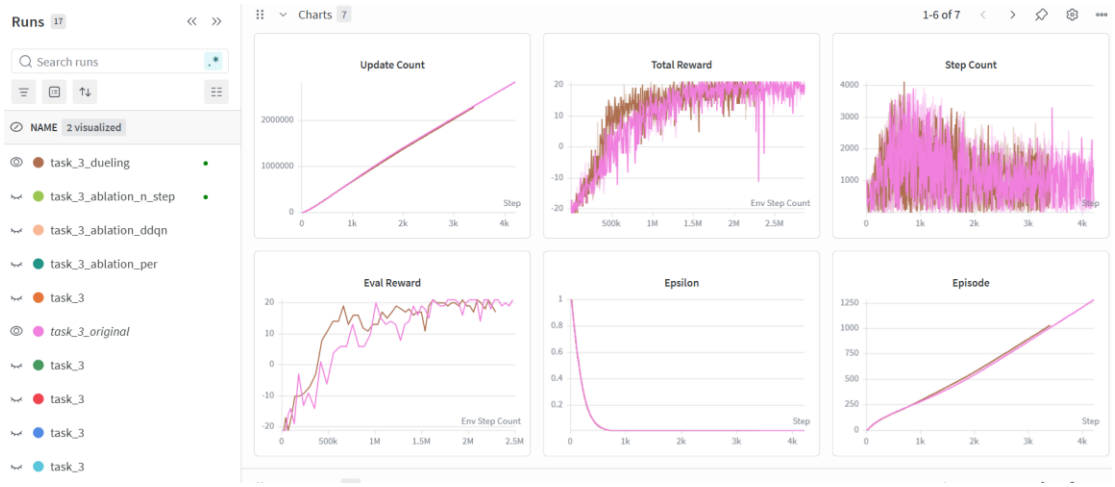
- How do you modify the 1-step return to multi-step return?

把多次的 reward 用 dequeue 存起來，然後套用公式累加起來。

```
if len(self.n_step_buffer) == self.n_step_n:
    R = sum(self.gamma ** i * self.n_step_buffer[i][2] for i in range(self.n_step_n))
    s0, a0, _ = self.n_step_buffer.popleft()
    self.memory.add((s0, a0, R, next_state, done), self.memory.max_priority)
```

- Explain how you use Weight & Bias to track the model performance.

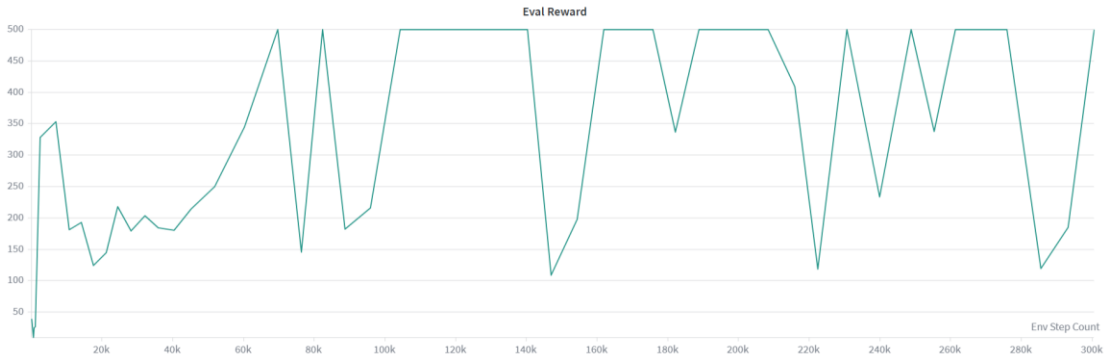
觀察學習曲線，找出最好的參數。



- Analysis and discussions (25%)

- Plot the training curves (evaluation score versus environment steps) for Task 1, Task 2, and Task 3 separately, please make sure the x-axis represent environment step (3+3+4=10%).

Task 1:快速達到滿分



Task 2:訓練非常多回合才達到平均 19 分。



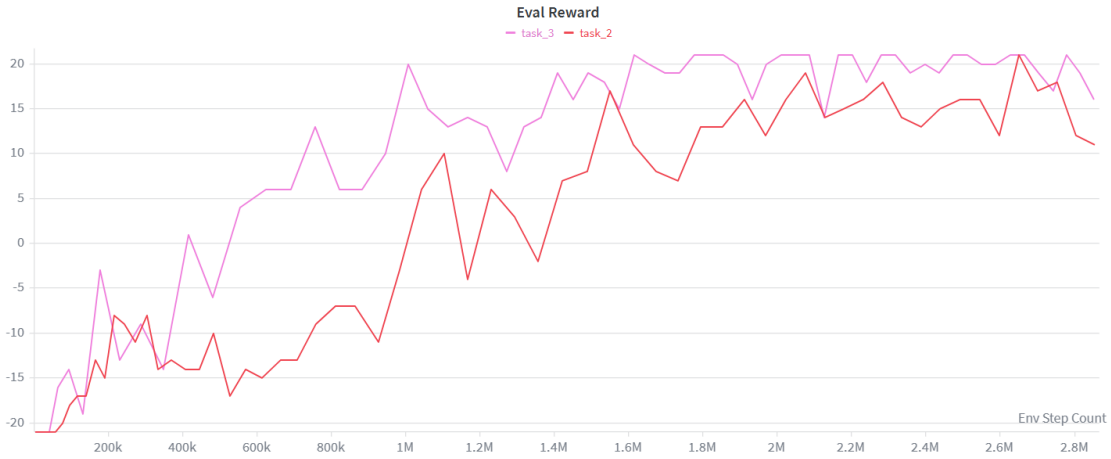
Task 3:較快的收斂進度



- Analyze the sample efficiency with and without the DQN enhancements (10%).

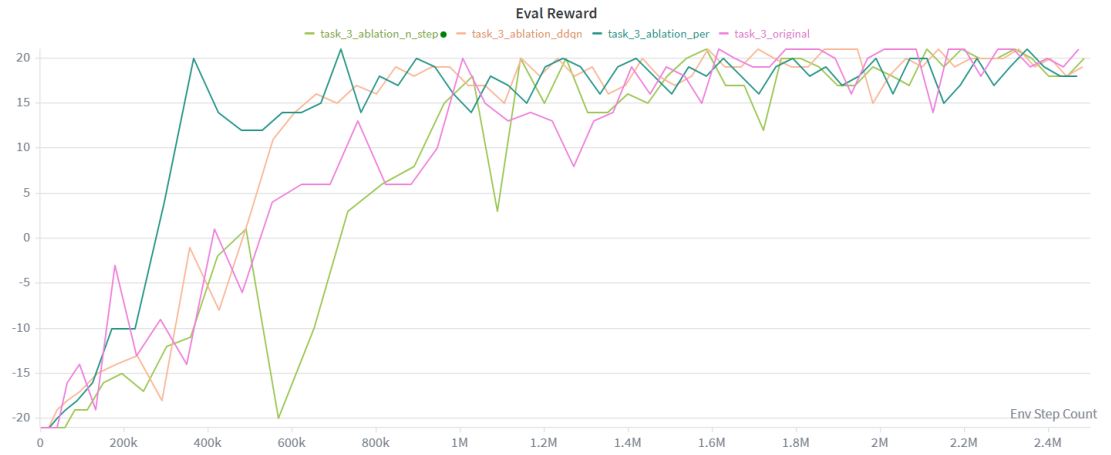
加上那些 enhancements 以後，模型不用再訓練到四百多萬個 env step 就有平

均 19 分了，整體而言效率提升非常多。



- Please perform an ablation study on each technique separately (5%).

	600000	1000000	1500000	2000000	2500000
Original	5.40	10.30	19.10	19.40	19.70
-PER	16.45	18.00	18.35	19.20	18.45
-DDQN	13.65	17.55	19.05	19.40	18.35
-N-step	-2.40	10.15	16.95	18.90	



結果顯示，拿掉 n-step 是最嚴重的，整體 eval 的分數最低。而拿掉 PER 雖然初始訓練較快，但後期的分數沒那麼高。而如果加上這三個項目，分數能達到最高分 19.7。

- Additional analysis on other training strategies (Bonus up to 10%)

我還有額外把神經網路的架構改成 Dueling DQN，是把原本的神經網路拆成價值分支和優勢分支，一個評估 state、一個評估 action，並把兩者的結果相加和處理，得到新的模型架構，整體而言收斂較快(較快達到高分)。

```python
class DuelingDQN(nn.Module):
    def __init__(self, input_channels, num_actions):
        super(DuelingDQN, self).__init__()
        self.features = nn.Sequential(
            nn.Conv2d(input_channels, 32, kernel_size=8, stride=4),
            nn.ReLU(),
            nn.Conv2d(32, 64, kernel_size=4, stride=2),
            nn.ReLU(),
            nn.Conv2d(64, 64, kernel_size=3, stride=1),
            nn.ReLU(),
            nn.Flatten()
        )
        self.value_stream = nn.Sequential(
            nn.Linear(64 * 7 * 7, 512),
            nn.ReLU(),
            nn.Linear(512, 1)
        )
        self.advantage_stream = nn.Sequential(
            nn.Linear(64 * 7 * 7, 512),
            nn.ReLU(),
            nn.Linear(512, num_actions)
        )

    def forward(self, x):
        features = self.features(x / 255.0)
        values = self.value_stream(features)
        advantages = self.advantage_stream(features)

        q_values = values + (advantages - advantages.mean(dim=1, keepdim=True))
        return q_values
```



Task 1:

```
(evan) C:\Users\user\Downloads\DL_LAB5>python test_model_task1.py --model-path ./LAB5_111550041_task1.pt
Seed: 0, eval reward: 500.0
Seed: 1, eval reward: 500.0
Seed: 2, eval reward: 500.0
Seed: 3, eval reward: 500.0
Seed: 4, eval reward: 500.0
Seed: 5, eval reward: 500.0
Seed: 6, eval reward: 500.0
Seed: 7, eval reward: 500.0
Seed: 8, eval reward: 500.0
Seed: 9, eval reward: 500.0
Seed: 10, eval reward: 500.0
Seed: 11, eval reward: 500.0
Seed: 12, eval reward: 500.0
Seed: 13, eval reward: 500.0
Seed: 14, eval reward: 500.0
Seed: 15, eval reward: 500.0
Seed: 16, eval reward: 500.0
Seed: 17, eval reward: 500.0
Seed: 18, eval reward: 500.0
Seed: 19, eval reward: 500.0
Average reward: 500.0
```

Task 2:

```
(evan) C:\Users\user\Downloads\DL_LAB5>python test_model.py --model-path ./results_pong/LAB5_111550041_task2.pt --env-steps 4272000 --seed 813
A.L.E: Arcade Learning Environment (version 0.10.2+c9d4b19)
[Powered by Stella]
Environment steps: 4272000, seed: 813, eval reward: 20.0
Environment steps: 4272000, seed: 814, eval reward: 19.0
Environment steps: 4272000, seed: 815, eval reward: 20.0
Environment steps: 4272000, seed: 816, eval reward: 19.0
Environment steps: 4272000, seed: 817, eval reward: 19.0
Environment steps: 4272000, seed: 818, eval reward: 19.0
Environment steps: 4272000, seed: 819, eval reward: 15.0
Environment steps: 4272000, seed: 820, eval reward: 18.0
Environment steps: 4272000, seed: 821, eval reward: 18.0
Environment steps: 4272000, seed: 822, eval reward: 21.0
Environment steps: 4272000, seed: 823, eval reward: 19.0
Environment steps: 4272000, seed: 824, eval reward: 17.0
Environment steps: 4272000, seed: 825, eval reward: 18.0
Environment steps: 4272000, seed: 826, eval reward: 21.0
Environment steps: 4272000, seed: 827, eval reward: 17.0
Environment steps: 4272000, seed: 828, eval reward: 19.0
Environment steps: 4272000, seed: 829, eval reward: 21.0
Environment steps: 4272000, seed: 830, eval reward: 19.0
Environment steps: 4272000, seed: 831, eval reward: 20.0
Environment steps: 4272000, seed: 832, eval reward: 21.0
Average reward: 19.0
```

Task 3:

600000

```
(evan) C:\Users\user\Downloads\DL_LAB5>python test_model.py --model-path ./results_pong/LAB5_111550041_task3_600000.pt --env-steps 600000
A.L.E: Arcade Learning Environment (version 0.10.2+c9d4b19)
[Powered by Stella]
Environment steps: 600000, seed: 313551076, eval reward: 5.0
Environment steps: 600000, seed: 313551077, eval reward: -2.0
Environment steps: 600000, seed: 313551078, eval reward: 3.0
Environment steps: 600000, seed: 313551079, eval reward: 3.0
Environment steps: 600000, seed: 313551080, eval reward: 1.0
Environment steps: 600000, seed: 313551081, eval reward: 8.0
Environment steps: 600000, seed: 313551082, eval reward: 7.0
Environment steps: 600000, seed: 313551083, eval reward: 7.0
Environment steps: 600000, seed: 313551084, eval reward: -2.0
Environment steps: 600000, seed: 313551085, eval reward: 9.0
Environment steps: 600000, seed: 313551086, eval reward: 4.0
Environment steps: 600000, seed: 313551087, eval reward: 12.0
Environment steps: 600000, seed: 313551088, eval reward: 11.0
Environment steps: 600000, seed: 313551089, eval reward: 1.0
Environment steps: 600000, seed: 313551090, eval reward: 11.0
Environment steps: 600000, seed: 313551091, eval reward: -5.0
Environment steps: 600000, seed: 313551092, eval reward: 6.0
Environment steps: 600000, seed: 313551093, eval reward: 14.0
Environment steps: 600000, seed: 313551094, eval reward: 10.0
Environment steps: 600000, seed: 313551095, eval reward: 5.0
Average reward: 5.4
```

1000000

```
(evan) C:\Users\user\Downloads\DL_LAB5>python test_model.py --model-path ./results_pong/LAB5_111550041_task3_1000000.pt --env-steps 1000000
A.L.E: Arcade Learning Environment (version 0.10.2+c9d4b19)
[Powered by Stella]
Environment steps: 1000000, seed: 313551076, eval reward: 15.0
Environment steps: 1000000, seed: 313551077, eval reward: 15.0
Environment steps: 1000000, seed: 313551078, eval reward: 9.0
Environment steps: 1000000, seed: 313551079, eval reward: 10.0
Environment steps: 1000000, seed: 313551080, eval reward: 7.0
Environment steps: 1000000, seed: 313551081, eval reward: 6.0
Environment steps: 1000000, seed: 313551082, eval reward: 4.0
Environment steps: 1000000, seed: 313551083, eval reward: 7.0
Environment steps: 1000000, seed: 313551084, eval reward: 8.0
Environment steps: 1000000, seed: 313551085, eval reward: 3.0
Environment steps: 1000000, seed: 313551086, eval reward: 9.0
Environment steps: 1000000, seed: 313551087, eval reward: 11.0
Environment steps: 1000000, seed: 313551088, eval reward: 12.0
Environment steps: 1000000, seed: 313551089, eval reward: 9.0
Environment steps: 1000000, seed: 313551090, eval reward: 11.0
Environment steps: 1000000, seed: 313551091, eval reward: 9.0
Environment steps: 1000000, seed: 313551092, eval reward: 13.0
Environment steps: 1000000, seed: 313551093, eval reward: 15.0
Environment steps: 1000000, seed: 313551094, eval reward: 16.0
Environment steps: 1000000, seed: 313551095, eval reward: 17.0
Average reward: 10.3
```

1500000

```
(evan) C:\Users\user\Downloads\DL_LAB5>python test_model.py --model-path ./results_pong/LAB5_111550041_task3_1500000.pt --env-steps 1500000
A.L.E: Arcade Learning Environment (version 0.10.2+c9d4b19)
[Powered by Stella]
Environment steps: 1500000, seed: 313551076, eval reward: 19.0
Environment steps: 1500000, seed: 313551077, eval reward: 18.0
Environment steps: 1500000, seed: 313551078, eval reward: 17.0
Environment steps: 1500000, seed: 313551079, eval reward: 18.0
Environment steps: 1500000, seed: 313551080, eval reward: 20.0
Environment steps: 1500000, seed: 313551081, eval reward: 19.0
Environment steps: 1500000, seed: 313551082, eval reward: 17.0
Environment steps: 1500000, seed: 313551083, eval reward: 20.0
Environment steps: 1500000, seed: 313551084, eval reward: 18.0
Environment steps: 1500000, seed: 313551085, eval reward: 21.0
Environment steps: 1500000, seed: 313551086, eval reward: 17.0
Environment steps: 1500000, seed: 313551087, eval reward: 21.0
Environment steps: 1500000, seed: 313551088, eval reward: 21.0
Environment steps: 1500000, seed: 313551089, eval reward: 21.0
Environment steps: 1500000, seed: 313551090, eval reward: 16.0
Environment steps: 1500000, seed: 313551091, eval reward: 19.0
Environment steps: 1500000, seed: 313551092, eval reward: 18.0
Environment steps: 1500000, seed: 313551093, eval reward: 21.0
Environment steps: 1500000, seed: 313551094, eval reward: 20.0
Environment steps: 1500000, seed: 313551095, eval reward: 21.0
Average reward: 19.1
```

2000000

```
(evan) C:\Users\user\Downloads\DL_LAB5>python test_model.py --model-path ./results_pong/LAB5_111550041_task3_2000000.pt --env-steps 2000000
A.L.E: Arcade Learning Environment (version 0.10.2+c9d4b19)
[Powered by Stella]
Environment steps: 2000000, seed: 313551076, eval reward: 20.0
Environment steps: 2000000, seed: 313551077, eval reward: 18.0
Environment steps: 2000000, seed: 313551078, eval reward: 20.0
Environment steps: 2000000, seed: 313551079, eval reward: 20.0
Environment steps: 2000000, seed: 313551080, eval reward: 21.0
Environment steps: 2000000, seed: 313551081, eval reward: 18.0
Environment steps: 2000000, seed: 313551082, eval reward: 20.0
Environment steps: 2000000, seed: 313551083, eval reward: 21.0
Environment steps: 2000000, seed: 313551084, eval reward: 15.0
Environment steps: 2000000, seed: 313551085, eval reward: 19.0
Environment steps: 2000000, seed: 313551086, eval reward: 21.0
Environment steps: 2000000, seed: 313551087, eval reward: 19.0
Environment steps: 2000000, seed: 313551088, eval reward: 21.0
Environment steps: 2000000, seed: 313551089, eval reward: 20.0
Environment steps: 2000000, seed: 313551090, eval reward: 21.0
Environment steps: 2000000, seed: 313551091, eval reward: 17.0
Environment steps: 2000000, seed: 313551092, eval reward: 19.0
Environment steps: 2000000, seed: 313551093, eval reward: 19.0
Environment steps: 2000000, seed: 313551094, eval reward: 19.0
Environment steps: 2000000, seed: 313551095, eval reward: 20.0
Average reward: 19.4
```

2500000

```
(evan) C:\Users\user\Downloads\DL_LAB5>python test_model.py --model-path ./results_pong/LAB5_111550041_task3_2500000.pt --env-steps 2500000
A.L.E: Arcade Learning Environment (version 0.10.2+c9d4b19)
[Powered by Stella]
Environment steps: 2500000, seed: 313551076, eval reward: 19.0
Environment steps: 2500000, seed: 313551077, eval reward: 15.0
Environment steps: 2500000, seed: 313551078, eval reward: 18.0
Environment steps: 2500000, seed: 313551079, eval reward: 20.0
Environment steps: 2500000, seed: 313551080, eval reward: 20.0
Environment steps: 2500000, seed: 313551081, eval reward: 17.0
Environment steps: 2500000, seed: 313551082, eval reward: 20.0
Environment steps: 2500000, seed: 313551083, eval reward: 21.0
Environment steps: 2500000, seed: 313551084, eval reward: 21.0
Environment steps: 2500000, seed: 313551085, eval reward: 20.0
Environment steps: 2500000, seed: 313551086, eval reward: 20.0
Environment steps: 2500000, seed: 313551087, eval reward: 20.0
Environment steps: 2500000, seed: 313551088, eval reward: 20.0
Environment steps: 2500000, seed: 313551089, eval reward: 21.0
Environment steps: 2500000, seed: 313551090, eval reward: 19.0
Environment steps: 2500000, seed: 313551091, eval reward: 21.0
Environment steps: 2500000, seed: 313551092, eval reward: 21.0
Environment steps: 2500000, seed: 313551093, eval reward: 20.0
Environment steps: 2500000, seed: 313551094, eval reward: 20.0
Environment steps: 2500000, seed: 313551095, eval reward: 21.0
Average reward: 19.7
```

```
(evan) C:\Users\user\Downloads\DL_LAB5>python test_model.py --model-path ./results_pong/LAB5_111550041_task3_best.pt --env-steps 1614457
A.L.E: Arcade Learning Environment (version 0.10.2+c9d4b19)
[Powered by Stella]
Environment steps: 1614457, seed: 313551076, eval reward: 21.0
Environment steps: 1614457, seed: 313551077, eval reward: 21.0
Environment steps: 1614457, seed: 313551078, eval reward: 18.0
Environment steps: 1614457, seed: 313551079, eval reward: 20.0
Environment steps: 1614457, seed: 313551080, eval reward: 20.0
Environment steps: 1614457, seed: 313551081, eval reward: 20.0
Environment steps: 1614457, seed: 313551082, eval reward: 20.0
Environment steps: 1614457, seed: 313551083, eval reward: 21.0
Environment steps: 1614457, seed: 313551084, eval reward: 21.0
Environment steps: 1614457, seed: 313551085, eval reward: 21.0
Environment steps: 1614457, seed: 313551086, eval reward: 19.0
Environment steps: 1614457, seed: 313551087, eval reward: 19.0
Environment steps: 1614457, seed: 313551088, eval reward: 20.0
Environment steps: 1614457, seed: 313551089, eval reward: 21.0
Environment steps: 1614457, seed: 313551090, eval reward: 20.0
Environment steps: 1614457, seed: 313551091, eval reward: 20.0
Environment steps: 1614457, seed: 313551092, eval reward: 20.0
Environment steps: 1614457, seed: 313551093, eval reward: 20.0
Environment steps: 1614457, seed: 313551094, eval reward: 20.0
Environment steps: 1614457, seed: 313551095, eval reward: 20.0
Average reward: 20.1
```